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Paper Code : BCAC601 Unix and Shell programming

UPID : 600105

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) Which command is used to display all the files including hidden files in your current and its sub directories?

(II) How do you add (append) a file “file1” to the “example.tar” file?

(III) Who can change the password of any user?

(IV) How do you get help about the command “cp”?

(V) Which command is used to extract a column from a text file?

(VI) Which command is used to identify file type?

(VII) Explain the file permission 764.

(VIII) The permission -rwxr–r– represented in octal expression will be ____.

(IX) What is the utility of ln command?

(X) What command is used to copy files and directories?

(XI) User’s Primary Group id is listed in which file, at the time of creation of the user (On a standard Unix system) is _____.

(XII) What is the user id of the administrator?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Differentiate between ‘user primary group’ and ‘user secondary group’ with suitable examples. [5]
3. Differentiate between orphan process and zombie process. What is the utility of ‘fork’ and ‘wait’ system calls? [5]
4. What is the way to change file and directory permission? Briefly describe with both techniques. [5]
5. Write down the short notes on the UNIX File System. [5]
6. What is the defunct and daemon process in UNIX? [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Write a shell script, which receives two file names as arguments, check whether the two files contents are the same or not and if they are same then the second file is deleted. [7]
   (b) Write a shell script to generate all combinations of 1, 2 and 3. [8]
8. (a) Explain Redirection in UNIX? [5]
   (b) Differentiate among cd, cd .. and cd – commands with proper examples. [5]
   (c) Explain grep –E with suitable examples. [5]
9. (a) Describe different modes of vi editor. [5]
   (b) Describe the two main functions of init. What is the significance of run level 0, 1 and 6? [2+3]
   (c) Explain the commands ls –l, ls –lt, ls –lu, ls –i, ls –a. [5]
10. (a) Explain boot block. [2]
    (b) What is signal? Give the names of five signals and their descriptions. [5]
    (c) Write a shell program for bubble sorting. [8]
11. (a) What is the difference between su, su - and su user1? [5]
    (b) Write a function that will generate and print first n Fibonacci numbers. [5]
    (c) What is the difference between the following vi commands? (i) :w (ii) :q! (iii) :wq (iv) zz (v) yy [5]